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All your doubts cleared in one place

Understanding Dolby Vision and Dolby Vision IQ

What makes them better than HDR10 and HDR10+?

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What if your TV could reveal every subtle nuance of a sunset, from the blazing sun to the delicate hues of twilight? This is the promise of Dolby Vision - one of the most popular HDR formats which is now available across TVs in affordable, mid-range and premium price segments.

But what exactly is HDR, and how does it work? How does Dolby Vision differ from regular HDR, and why does it matter in your home theatre setup? What are the factors you need to look at for the best Dolby Vision experience? Here we will try to understand what Dolby Vision is in detail and answer all related questions that you might have.

Dolby Vision vs HDR10

Now, the problem is that TVs don't have the same capabilities as the mastering monitors that cost several $1000 dollars. Most consumer TVs struggle to hit 1000 nits of peak brightness and might not support as wide a colour gamut. So, to do justice to the HDR content, the TV tone maps the brightness, contrast and colours within its individual display characteristics.

Let's say you have a mastering monitor with a peak brightness of 1000 nits, and you're working on a scene with a sun that's supposed to be at that peak brightness. The sun is the brightest object in the scene, with a value of 1000 nits. If your TV supports a maximum of 500 nits of peak brightness, the tone mapping algorithm reduces the brightness of the sun from 1000 nits to 500 nits, to match the TV's peak brightness. The algorithm will also adjust the brightness of the surrounding areas, like the sky and landscape, to maintain a consistent contrast ratio.

This tone mapping can make or break the HDR experience on your TV. Manufacturers have to choose between preserving highlight details or details in bright scenes and maintaining a bright Average Picture Level bright APL (Average Picture Level) or keeping the overall image bright and punchy.

This information regarding the primaries of the mastering monitor is conveyed to the TV via Metadata. HDR10 has static metadata which specifies the maximum brightness of the entire content (MaxCLL) and the average brightness of the brightest frame (MaxFLL) in the content. This means the entire tone mapping for a movie that you are watching will be defined by these two values. Dolby Vision, on the other hand, has dynamic metadata and can change these values several times for content. In fact, content creators can specify light level values on a scene-by-scene (shot-based) or frame-by-frame (frame-based) basis! If graded properly, this results in a much more refined HDR experience as your TV can tone map bright scenes and dark scenes differently.

Apart from light levels, Dolby Vision metadata carries a range of additional parameters that can be used for more detailed control. Colourists have the option to create multiple 'trim passes' during post-processing where they can manually configure metadata to help content easily adapt to different target displays while staying as true to creative intent as possible.

This is essentially the main difference between Dolby Vision and Regular HDR10. HDR10+ is another format that offers dynamic metadata but Dolby Vision is by far more popular and HDR10+ content remains